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CONTROLLED POROSITY OF CERAMIC SCAFFOLD BY DIRECTIONAL FREEZE CASTING AND SCAFFOLD PRINTING

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ABSTRACT

The objective of this work is to produce a 3-dimensional scaffold with hierarchical porosity produced with automated manufacturing control. In this paper we present a two-step manufacturing process: solid freeform fabrication (SFF) of a polymer scaffold, followed by freeze casting, a directional solidification technique, of composite slurry. This combination allows for the production of composite scaffolds with well defined, gradient porosity in two directions. Transverse porosity is created by immersing a printed alginate scaffold in a chitosan (CS)-hydroxyapatite (HA) slurry which is freeze cast to produce a scaffold with highly aligned porosity in the longitudinal direction. The combination of these two processes, printing and freeze casting, provides engineering control over chemical, structural and mechanical cues of the interconnected microenvironment within the scaffolds over several length scales, including features like overall porosity, pore size and shape as well as the choice of material and for example the ratio of polymer to ceramic within the highly porous composite

material. Thus, the material's structure and architecture can be custom-designed by the fabrication processes to regulate diffusion throughout the scaffold as well as to potentially direct cell proliferation and migration.

1 INTRODUCTION

Scaffold architecture is a cue known to direct cell behavior in a functionalized tissue engineered product [1-4]. Desirable features for *in vitro* culture environments are derived from native tissue and parametric studies of cell response to engineered scaffolds [5, 6]. The microarchitecture, such as pore size, geometry, and interconnectivity is tailored for maximum cell infiltration [7, 8] and to produce substrate and diffusion conditions to promote specific cell behavior and function [9-11]. The manufacturability and reproducibility of the microarchitecture depends on the control of independent scaffold properties [12, 13]. Gradient and hierarchical structures within 3-dimensional scaffolds can selectively

stimulate cell behavior through structural and mechanical cues [14-17].

The objective of this work is to produce a 3-dimensional scaffold with hierarchical porosity with an automated process. The combination of two fabrication processes offers unique engineering control of pore size and material distribution of the ceramic and polymer phase in three mutually perpendicular directions to produce hierarchical structured microenvironments for scaffold-guided tissue engineering. In this paper, we present a two-step manufacturing process, in which ceramic-polymer slurry is freeze cast around a porous alginate scaffolds that is embedded in it. After freeze casting, the two scaffolds present hierarchical microenvironment niches: the transverse SFF alginate struts are embedded in the lamellar structure of the freeze-cast polymer-mineral composite. This composite scaffold architecture possesses hierarchical porosity in mutually perpendicular directions; longitudinal porosity due to unidirectional freezing and transverse porosity created by the embedded alginate scaffold.

Automated manufacturing process parameters allow for an engineering control of the composite scaffold microarchitecture. The alginate scaffold is printed using a multi-nozzle bioprinter system for the controlled deposition of alginate and a cross-linker to simultaneously print and cross-link alginate. The alginate ligaments are printed by continuous deposition through micron-scale capillary tips to form 2-dimensional planar layers, which are stacked to produce a 3-dimensional scaffold. The pattern of printed alginate is defined by design model inputs using automation software. The diameter of the printed filaments is controlled by the flow rate through the nozzle as defined in the automation software. The porous alginate scaffold is immersed in the ceramic-polymer slurry under vacuum and placed in a mold consisting of a PTFE tube and a copper bottom plate prior to freeze casting. The cooling rate is set to 10°C/min and controlled by a band heater and a liquid nitrogen bath acting through a copper cold finger. The cooling rate dictates the local cooling rate and thereby the lamellar spacing in the structure, which defines the size of the lamellar pores. After complete solidification, the frozen composite is lyophilized to sublimate the ice and produce

porosity within the alginate filaments and the ceramic-polymer composite.

The combination of manufacturing processes described above provides engineering control over the hierarchical microenvironment within the composite scaffolds from the nanometer to the centimeter length scales. The printing and freeze casting process enables independent process control of the polymer to ceramic ratio as well as the structure and architecture of the porous material. The hierarchical porosity of the scaffold can carefully be controlled by two independent process parameters (1) the flow rate during scaffold printing (2) the freezing rate during freeze casting. Both parameters can be continuously adjusted during printing/freezing to produce gradient or hierarchical structures. This approach, therefore, has great promise for scaffold-guided tissue engineering, 3-dimensional co-culture, and parametric studies of the microenvironment's effect on cell behavior due to the processes' ability to create custom-designed scaffold features.

2 MATERIALS AND METHODS

2.1 SFF Multi-Nozzle Bioprinter System

SFF of biological material and matrix by direct cell writing produces well-defined 3-dimensional patterns by layer-by-layer deposition [18, 19]. Multi-nozzle printing provides the means for independent automated control of alginate and cross-linker deposition for simultaneous scaffold printing and cross-linking. Sodium alginate (6% w/v with distilled water) and cross-linker (5% calcium chloride w/v in distilled water) are loaded into separate pneumatic microvalves and mixed upon deposition during printing to produce a rigid hydrogel scaffold. Figure 1 presents a schematic of the bioprinter and multi-nozzle setup. The material delivery system is mounted on a high precision positioning system to print alginate scaffold layer-by-layer to produce a 3-dimensional scaffold. Design model inputs and manufacturing process parameters, excluding dispensing capillary tip diameter, are defined by the operator through automation software Motion Planner (Parker Hannifin Corp). The nozzle is selected from a pre-fabricated set of precision capillary tips and fixed to the material delivery system prior to extrusion. The material delivery system feeds alginate matrix through an interchangeable micron-scale capillary tip by continuous pneumatic pressure regulated from reservoir of breathing quality compressed air (AirGas). Once pressurized, the automation software is run to extrude with 15 psi of air pressure the alginate matrix through a 250 μm capillary tip that is 12.7 mm in length. The capillary tip is positioned 0.25 mm above the substrate and moved upward by 0.25 mm between layers. The alginate scaffold is printed in a square wave pattern, with outside dimensions of 20 mm $\times$ 20 mm with 2.0 mm center-to-center filament spacing. Subsequent layers are mutually orthogonal.

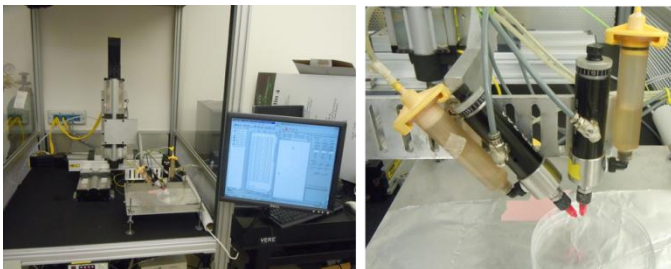


Fig. 1. Bioprinter system (left) and multi-nozzle delivery system set-up for controlled deposition of alginate and cross-linker for simultaneous scaffold printing and cross-linking (right).

2.2 Process of Freeze Casting Composite Slurry

The resulting porous alginate scaffolds are immersed, under vacuum, in a HA-CS slurry for directional solidification in a cylindrical mold that is sealed by a copper bottom plate for directional freezing atop a cold finger, which is immersed in a liquid nitrogen bath. The cooling rate applied by the cold finger top is regulated with a PID-controller that is attached to a thermocouple at the top of the cold finger and a band heater just below it to counteract the liquid nitrogen cooling bath [20]. The applied freezing rate, which can be carefully controlled, determines the sample cooling rate within the mold, and through it the lamellar spacing within the freeze-cast scaffold. After freeze casting, the sample is lyophilized to sublime the ice-phase which templates porosity within both the printed alginate and the freeze-cast ceramic scaffolds. Finally, the sample is sintered at a temperature of 1250°C for 1 hour to remove the alginate filaments as well as the CS binder. The result is a ceramic scaffold with ice-templated longitudinal and alginate-templated transverse pores.

3 RESULTS AND DISCUSSION

3.1 Porous Alginate Scaffold Produced by SFF Multi-Nozzle Bioprinting

To produce the transverse porosity in the freeze-cast scaffold, a porous alginate scaffold is printed using multi-nozzle bioprinting and immersed in a CS-HA slurry. Sodium alginate

and the cross-linking agent calcium chloride are loaded into independent pneumatic microvalves. Design model is programmed to a 2.0 mm center-to-center square wave filament spacing.

Complete scaffolds have 20 layers, where consecutive layers are mutually orthogonal (0°-90° orientation), with 0.2 mm elevation change between layers. A 250 μm capillary tip is attached to each microvalve. The travel velocity of both pneumatic microvalves is set to 5 mm/s and dispensing pressure is 15 psi for alginate and 3 psi for calcium chloride. Microvalves are programmed to extrude concurrently. Cross-linking occurs upon deposition of alginate to form scaffold. A printed scaffold is presented in Figure 3.

Alginate scaffolds are measured and photographed using phase contrast microscopy and image analysis software. The center-to-center filament spacing is 1.96 mm and the average filament diameter is 275 μm . The center-to-center spacing in the printed scaffold varied 1.8% from the programmed value and design model input. The filament diameter increased 10% from the capillary tip. Phase contrast images are presented in Figure 4. Before freeze casting, the alginate scaffolds were immersed in cross-linking solution and stored in a 100% humidity cabinet for 2 hours. For freeze casting, the cross-linking solution was drained and the alginate scaffold immersed in the HA/CS slurry. The composite slurry was degassed under vacuum for 5-10 minutes immediately prior to directional freezing.

3.2 Composite Scaffold with Hierarchical Microarchitecture Produced by Freeze Casting

To produce a hierarchical microarchitecture, the composite slurry with the immersed SFF alginate scaffold is directionally solidified and subsequently lyophilized. During the directional solidification, a phase separation occurs within the slurry into lamellae of pure ice and lamellae of a polymer-HA composite. These lamellae form parallel to the applied temperature gradient and thus normal to the bottom copper plate, which is the source of cooling. After complete solidification, the specimens are transferred to a lyophilizer to remove the ice within the structure and thus create the desired porosity.

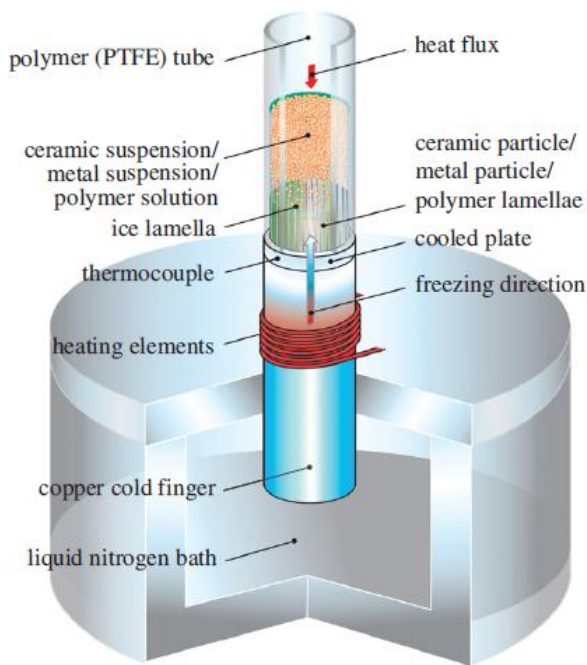


Fig. 2. Schematic of freeze-casting system, reproduced from Wegst et. al. [20]

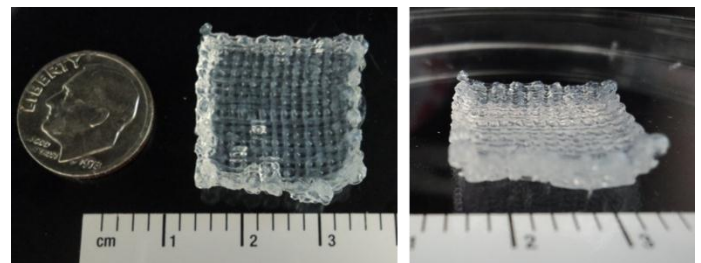


Fig. 3. 20-layer sodium alginate scaffold cross-linked upon deposition by multinozzle bioprinter.

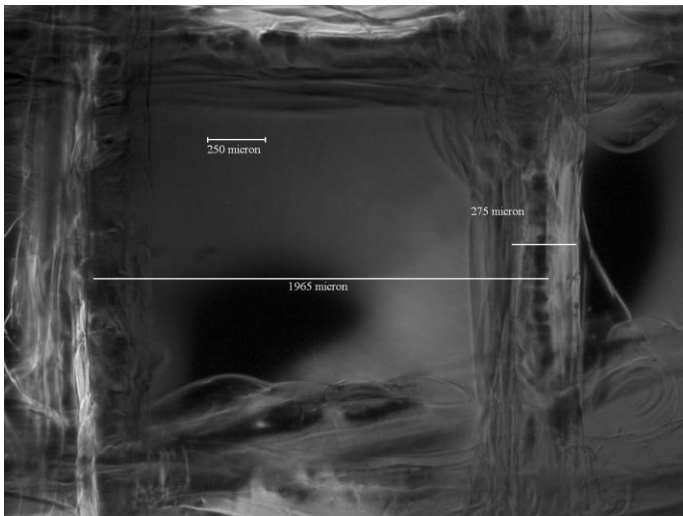


Fig. 4. Sodium alginate scaffold printed through $\text{Ø}250\mu\text{m}$ at 15psi dispensing pressure with traveling speed of 5 mm/s and 0.2mm elevation change between layers and 2000 μm center-to-center filament spacing.

The alginate filaments remain embedded in the polymer-ceramic scaffold after freeze drying but shrink significantly in diameter and hence produce regular porosity traversing that between the lamellae. A scanning electron micrograph of a longitudinal section through the composite scaffold is shown in Figure 6. The tubular pores created by the lateral alginate filaments extend in the direction normal to the cross-section. Their arrangement in rows and columns follows a regular pattern, templated by the SFF scaffold. This pattern of the embedded filaments is highlighted in red in Figure 5. The resulting material has a hierarchical porosity produced by both the SFF scaffold and the freeze casting process. This is further illustrated in the longitudinal cross-section of Figure 6, where alginate filaments extend out of the freeze-cast composite. These alginate filaments were deflected downwards during the sectioning process.

3.3 Ceramic Freeze Cast Scaffold with Hierarchical Microarchitecture

Purely ceramic scaffolds with increased stiffness and strength can be prepared when the freeze-cast composite scaffolds are sintered. During this sintering step, all organic matter within the scaffold is completely removed. An SEM image of a longitudinal section of the composite scaffold after sintering is shown in Figure 7. The transverse porosity, template by the FCC filaments as well as the longitudinal porosity of the freeze casting process are maintained and a hierarchically-structured ceramic scaffold fabricated. Figure 8 shows SEM images of the wall material before and after sintering. They reveal a third-level, submicron porosity within the lamellar pore walls, which allows for additional lateral diffusion through the scaffolds.

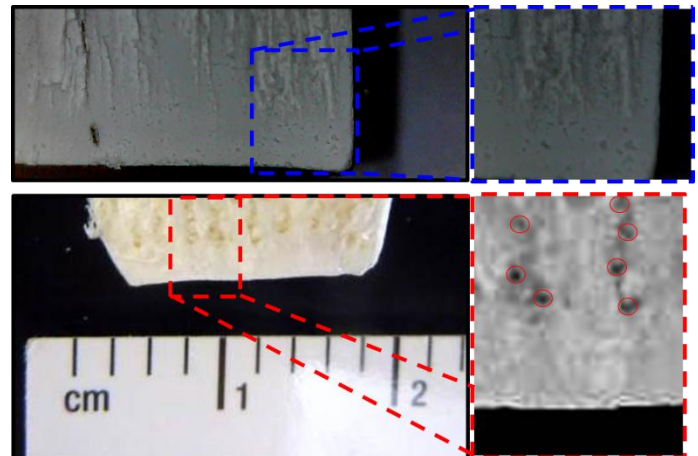


Fig. 5. (Top Row) Pure ceramic and (Bottom Row) composite of ceramic and porous alginate scaffold after freeze casting and then sectioned longitudinally. Red circles in composite indicate lateral porosity produced by embedded alginate scaffold.



Fig. 6. SEM images of pores created by alginate filaments embedded in ceramic slurry. Alginate filaments still occupy pores before sintering as shown. Alginate filament is visible pulled out of pore laying across image.

3.4 3-Dimensional Network of Interconnected Pores

The lateral pores, template by the SFF ligaments, connect longitudinal, lamellar pores, created during freeze casting, to produce interconnected hierarchical porosity, as shown in Figure 9. This 3-dimensional network has interconnected porosity at three levels of its structural hierarchy: 1) a lamellar, longitudinal porosity created by the

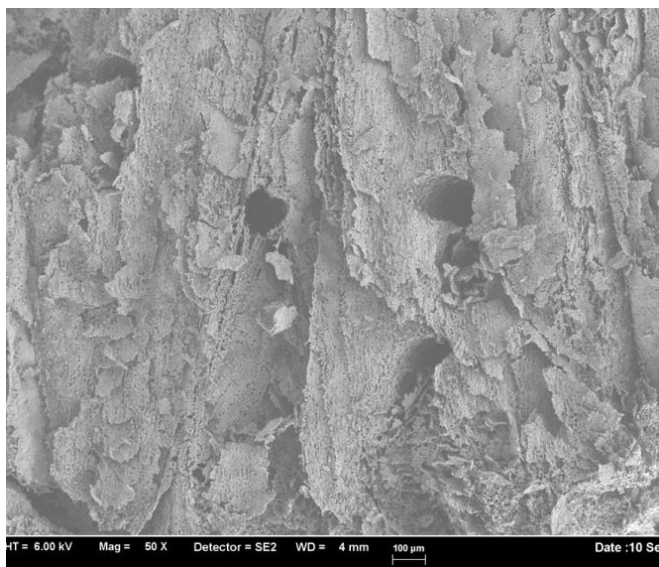


Fig. 7. SEM images of pores created by alginate filaments embedded in ceramic slurry after sintering.

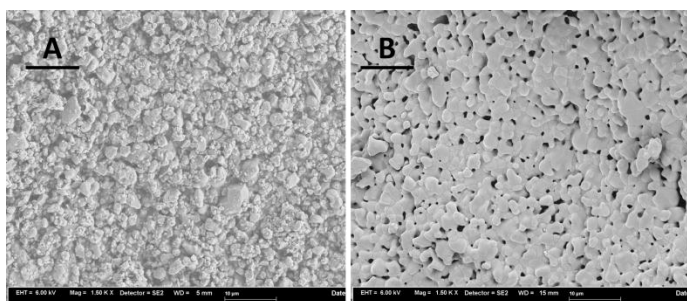


Fig. 8. SEM images ceramic before (A) and after (B) sintering.

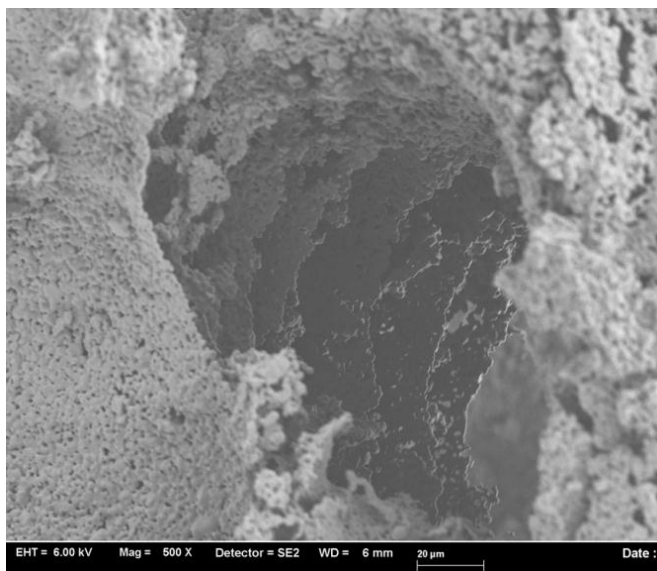


Fig. 9. SEM images of pores created by alginate filaments embedded in ceramic slurry. Alginate removed by sintering.

growing ice crystals during directional solidification 2) a tubular, transverse porosity templated by the ligaments of the SFF scaffold and 3) a transverse porosity within the lamellae in between adjacent ceramic grains, due to incomplete densification during sintering.

3.5 Post-Processed Material Allows is Biocompatible and Cell Attachment Occurs

Having established longitudinal and lateral porosity, we demonstrate the feasibility of a hierarchical ceramic scaffold to support cell adhesion. The freeze-cast HA-CS scaffolds are biocompatible and allow for cell attachment. The scaffolds used for cell culture are not sintered. Osteoprogenitor cells of the 7F2 cell line are seeded onto freeze cast printed alginate embedded in-chitosan-hydroxyapatite. Scaffolds were not sintered. Osteoprogenitor cells are chosen because one target tissue for a load bearing scaffold with lateral porosity is bone. Lateral porosity is desired to improve diffusion and possibly provide a microenvironment for angiogenesis. Cell survival and adhesion is observed by Live/Dead staining 24 hours after seeding. Cells were seeded on alginate-chitosan-hydroxyapatite, chitosan-hydroxyapatite, and alginate after freeze-casting. Figure 10 presents live cells (green) attached to the freeze-cast samples and control.

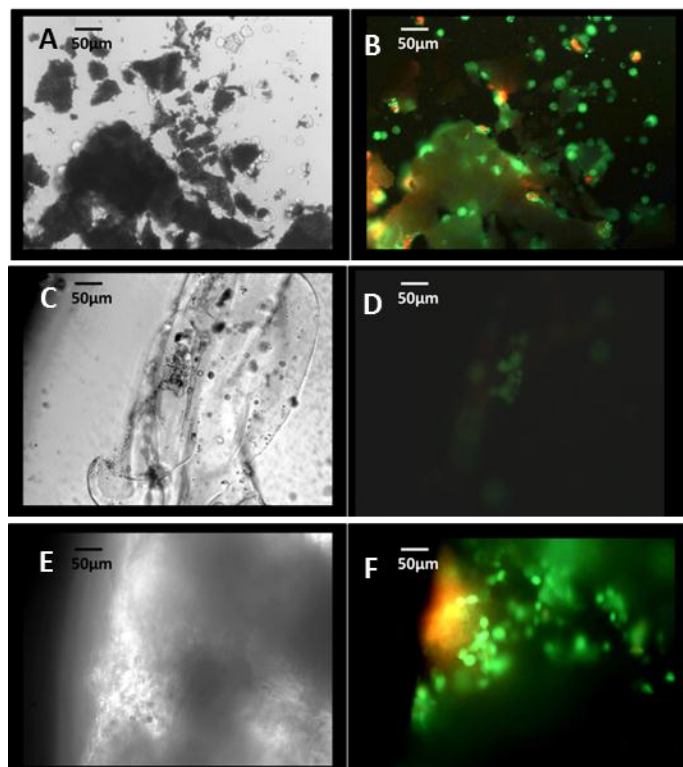


Fig. 10. 7F2 cells seeded on freeze cast (un-sintered) scaffold of chitosan-hydroxyapatite (A,B), alginate (C,D), and alginate-chitosan-hydroxyapatite (E,F) using phase contrast light microscope (left) and fluorescent (right) microscopy.

Viable cells are observed on all substrates. Limited viable cells are observed on the freeze cast alginate. Alginate itself does not interact with cells or allow cell anchorage. Significantly more viable cells are observed on the freeze-cast ceramic, as the addition of HA facilitates cell adhesion and material-cell interaction. The freeze-cast composite had significantly more viable cells than the pure alginate and slightly less than the pure ceramic. The composite's cell attachment is expected to be slightly less than that of the pure ceramic, as part of the composite material is alginate. Further experimentation is required to quantitatively compare the three environments for cell attachment. However, the composite material is biocompatible and cell attachment occurs.

4 CONCLUSION

A combination of SFF and freeze casting facilitates the fabrication of composite scaffolds with hierarchical porosity for lateral diffusion. We demonstrate the successful production of hierarchical porosity caused by the embedded SFF alginate scaffold and the freeze casting process using scanning electron microscope (SEM) images of cross-sectioned scaffold. Pores in ceramic are produced by embedded alginate filaments. Filament diameter, and therefore the pore diameter, is controlled by the scaffold printing and manufacturing process parameters and can be patterned throughout the scaffold. The freeze casting process produces longitudinal pores through the ceramic normal to the cooling plate; visible as lamellae. The size and orientation of these pores is controlled by cooling rate. This work towards composite scaffold and process controlled gradient porosity has great potential for the application as guiding scaffold in tissue engineering at, for example, the bone-cartilage interface.

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